

Office Hours: 10:00a - 10:50a
12:00p - 1:00p MSB 2117

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Solving systems of equations

$$\begin{cases} f(x,y)=0 \\ g(x,y)=0 \end{cases} \text{ solve for } x,y$$

$$\begin{aligned} 3x + 2y &= 5 \Rightarrow y = \frac{1}{2}(5 - 3x) = \frac{1}{2}(5 - 3 \cdot (-2)) \\ 7x + 4y &= 8 \end{aligned}$$

$= \frac{1}{2}(5 + 6)$
 $y = \frac{11}{2}$

$$7x + 2(5 - 3x) = 8$$

$$7x + 10 - 6x = 8$$

$$x = -2$$

$$(x-1)(x-3)=0$$

$$x-1=0 \quad x-3=0$$

$$x=1 \quad x=3$$

$$\left. \begin{array}{l} x^2-3x=0 \\ x(x-3)=0 \end{array} \right\}$$

$$x=0 \quad x-3=0$$

$$x=3$$

~~$$\frac{x(x-3)=0}{x \quad x}$$~~

~~$$x-3=0$$~~

~~$$x=3$$~~

~~$$\frac{x(x-3)=0}{x-3 \quad x-3}$$~~

~~$$x=0$$~~

$$4x(x^2+y^2-1)=0$$

$$4x=0$$

$$x^2+y^2-1=0$$

$$y=\pm 1$$

$$(0,1)$$

$$(0,-1)$$

$$(1,0)$$

$$(-1,0)$$

$$x=\pm 1$$

$$(1,1)$$

$$3xy(5-3x)=0$$

$$4x(6-2y)=0$$

$$3xy(5-3y-2x)=0$$

$$4x(6-2y)=0$$

$$\frac{x=0}{(1)}$$

$$\frac{y=0}{(2)}$$

$$\frac{3y+2x=5}{(3)}$$

$$(1) x=0 \Rightarrow 4x(6-2y)=0$$

$$(0, 3)$$

$$4x=0$$

$$6-2y=0$$

$$y=3$$

$$(2) y=0 \Rightarrow 4x=0$$

$$x=0$$

$$\text{or } 6-2y=0$$

$$6-2 \cdot 0=0$$

$$y \neq 0$$

$$(0, 0)$$

$$(3) 3y+2x=5$$

$$y = \frac{1}{3}(5-2x)$$

$$y = \frac{1}{3}(5-2 \cdot 0) = \frac{5}{3}$$

$$y = \frac{1}{3}(5-2 \cdot 2) = \frac{1}{3}$$

$$4x=0$$

$$x=0$$

$$6-2\left(\frac{1}{3}(5-2x)\right)=0$$

$$6-\frac{2}{3}(5-2x)=0$$

$$6-\frac{10}{3}+\frac{4}{3}x=0$$

$$x=2$$

$$\begin{matrix} (0, \frac{5}{3}) \\ (2, \frac{1}{3}) \end{matrix}$$

$$\textcircled{3} \quad 3y + 2x = 5$$

$$y = \frac{1}{3}(5 - 2x)$$

$$y = \frac{1}{3}(5 - 2 \cdot 0) = \frac{5}{3}$$

$$y = \frac{1}{3}(5 + 2 \cdot 2) = 3$$

$$4x = 0$$

$$x = 0$$

$$6 - 2\left(\frac{1}{3}(5 - 2x)\right) = 0$$

$$6 - \frac{2}{3}(5 - 2x) = 0$$

$$6 - \frac{10}{3} + \frac{4}{3}x = 0$$

$$x = -2$$

$$\begin{pmatrix} 0, \frac{5}{3} \\ -2, 3 \end{pmatrix}$$

$$(0, 0) \quad (0, 3) \quad (0, \frac{5}{3}) \quad (-2, 3)$$

$$3xy(5 - 3y - 2x) = 0$$

$$4x(6 - 2y) = 0$$

$$x = 0$$

$$\{(x, y) \in \mathbb{R}^2 \mid x = 0\}$$

$$3y(5-3y-2x)=0$$

$$y=0$$

$$y = \frac{1}{3}(5-2x)$$

$$\underline{y=0}$$

$$4x(6-2y)=0$$

$$4x(6-0)$$

$$x=0$$

$$(0,0), (-2,3)$$

$$\underline{4x(6-2y)=0}$$

$$\underline{y = \frac{1}{3}(5-2x)}$$

$$4x=0$$

$$x=0$$

$$(0, \frac{5}{3})$$

$$6-2(\frac{1}{3}(5-2x))=0$$

$$6-\frac{2}{3}(5-2x)=0$$

$$6-\frac{10}{3}+\frac{4}{3}x=0$$

$$x=-2$$

$$y=3$$

$$3y(5-3y-2x)=0$$

$$\underline{4x(6-2y)=0}$$

$$4x=0 \Rightarrow x=0$$

$$6-2y=0 \Rightarrow y=3$$

$$\textcircled{1} \underline{x=0}$$

$$3y(5-3y-2 \cdot 0)=0$$

$$15y-9y^2=0$$

$$y(15-9y)=0$$

$$y=0, y=\frac{5}{3}$$

$$\boxed{(0,0)} \quad \boxed{(0,\frac{5}{3})}$$

$$\textcircled{2} \underline{y=3}$$

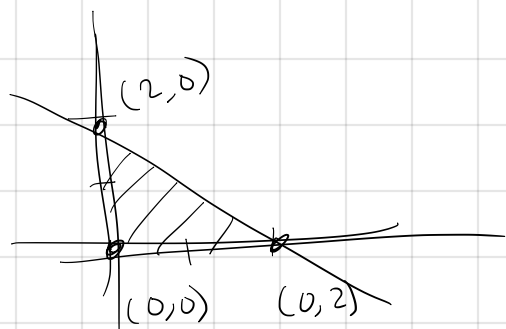
$$3 \cdot 3(5-3 \cdot 3-2x)=0$$

$$-4 \underbrace{5-9}-2x=0$$

$$x=-2$$

$$\boxed{(-2,3)}$$

$$f(x, y) = 2xy - x^2y - xy^2$$



$$x + y = 2$$

$$x\text{-axis} \Rightarrow y = 0$$

$$y\text{-axis} \Rightarrow x = 0$$

$$\nabla f(x, y) = \begin{bmatrix} 2y - 2xy - y^2 \\ 2x - x^2 - 2xy \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x - x^2 - 2x(2x - 2) = 0$$

$$2x - x^2 - 4x^2 + 4x = 0$$

$$-5x^2 + 6x = 0$$

$$-x(5x - 6) = 0$$

$$x = 0 \quad x = \frac{6}{5}$$

$$2y - 2xy - y^2 = 0$$

$$y(2 - 2x - y) = 0$$

$$\underline{y = 0}$$

$$2 - 2x - y = 0$$

$$\underline{y = 2x - 2}$$

$$2x - x^2 - 2xy = 0$$

$$\underline{y = 0} \quad 2x - x^2 = 0$$

$$x(2 - x) = 0$$

$$x = 0, \quad x = 2$$

$$\boxed{(0, 0)} \quad \boxed{(2, 0)}$$

~~$$(0, 2)$$~~

$$\left(\frac{6}{5}, \frac{2}{5} \right)$$

$$\underline{y = 2x - 2}$$

$$2 \cdot \frac{6}{5} - 2$$

$$\frac{2}{5}$$

not in
region

along edges

$$x=0 \Rightarrow f(0,y) = g_1(y)$$

$$g_1(y) = 2 \cdot 0 \cdot y - 0^2 y - 0y^2 \\ = 0$$

entire boundary are critical points

$$x+y=2 \Rightarrow y=2-x$$

$$g_3(x) = 2x(2-x) - x^2(2-x) - x(2-x)^2 \\ = 4x - 2x^2 - 2x^2 + x^3 - x(4 - 4x + x^2) \\ = 0$$

entire boundary are critical points

$$y=0$$

$$g_2(y) = 0 - 0 - 0 = 0$$

entire boundary are critical points

critical	$f(x,y)$	
$(0,0)$	0	
$(2,0)$	0	
$(0,2)$	0	
$x=0$	0	
$y=0$	0	
$x+y=2$	0	
$(\frac{6}{5}, \frac{2}{5})$	$\frac{24}{125}$	max

$$2 \cdot \frac{6}{5} \cdot \frac{2}{5} - \frac{6^2}{5^2} \cdot \frac{2}{5} - \frac{6}{5} \cdot \frac{2^2}{5^2}$$

$$\frac{24}{125} - \frac{72}{125} - \frac{24}{125}$$

$$\frac{120}{125} - \frac{72}{125} - \frac{24}{125} = \frac{24}{125}$$